

Credit Card Fraud Detection: End-to-End ML System Analysis and Redesign

AIML-500: Machine Learning Fundamentals · Group 3

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Assumption. No dataset, business thresholds, or regulatory jurisdiction was specified. Data sources, validation windows, and performance targets below are design proposals to be confirmed during implementation.

1. Problem Definition and Stakeholder Analysis

The system identifies fraudulent credit card transactions in real time while minimizing disruption to legitimate customers. It is a supervised binary classification problem: the target variable is 1 for a fraudulent transaction and 0 for a legitimate one. Rather than a single approve/decline output, the model produces a calibrated risk score that routes each transaction to one of four actions: approve, request additional authentication, send to a fraud analyst, or decline and hold.

Candidate predictors include transaction amount, timestamp, merchant and merchant category, location, device, channel, transaction velocity, and deviation from the cardholder's historical pattern.

Three properties of fraud shape every later decision in this report. Fraud is a small minority class, often under 0.1% of transactions. Fraud patterns change as criminals adapt, so a model degrades with time even when nothing about it changes. And labels are delayed, since a transaction is not confirmed as fraud until a dispute, chargeback, or investigation concludes.

Table 1

Stakeholders and Required System Outcomes

Stakeholder	Primary concern	Required outcome
Cardholders	Protection without unnecessary declines	Low false-positive and false-decline rates
Banks and issuers	Fraud-loss reduction	High fraud capture at positive net financial impact
Fraud analysts	Manageable, actionable alerts	High precision at realistic alert volumes
Merchants	Fast, reliable authorization	Low decision latency, minimal disruption
Customer-service teams	Fewer avoidable complaints	Reduced decline-related contacts and disputes
Compliance teams	Fairness, explainability, privacy	Documented controls, subgroup testing, human oversight
ML developers	Reproducible evaluation	Leakage-free data, time-aware validation, versioning

Success is measured on a balanced set of indicators rather than accuracy: precision, recall, PR-AUC, false-positive and false-decline rates, fraud dollars prevented, alert volume against analyst capacity, decision latency, score calibration, customer complaints, and consistency across segments. Overall accuracy is explicitly rejected as the headline metric, since a model that approves nearly everything scores above 99% accurate while catching almost no fraud.

2. Data Audit and Preprocessing Evaluation

The audit takes the position of a card issuer, the bank deciding in real time whether to approve or decline. That vantage point determines what data exists: an issuer sees every one of its cardholders but only its own cards, and merchants are a data source rather than a data owner.

Two constraints run through the whole pipeline. The authorization round trip is budgeted in the low hundreds of milliseconds, so a feature that cannot be computed inside that window is unusable regardless of offline performance. And the label is not observed at decision time. A Visa card-absent fraud dispute can be filed up to 120 days after the transaction (Chargebacks911, 2026), so recent training data always carries labels that are still filling in.

Table 2***Data Sources for an Issuer-Side Fraud Detection System***

#	Source	Contents	Latency	Data quality issues
1	Authorization stream	ISO 8583 fields: tokenized PAN, amount, currency, MCC, merchant and terminal ID, entry mode, AVS and CVV response, timestamps	Real time	Merchant descriptors unnormalized; MCC self-reported and often wrong; entry mode sometimes misreported
2	Posted ledger	Settled amounts, refunds, reversals, recurring billing	Daily batch	Settled differs from authorized (tips, fuel holds); reversals duplicate spend if not paired
3	Account master	Tenure, product, limit, home postal code, card age, reissue history	Daily	Reissue cuts a card off from its own history; demographic fields sit here and must not reach the model
4	Disputes, chargebacks, TC40 and SAFE	Fraud claims, chargeback reason codes, network fraud reports. This is the label	30 to 120 days	Late, partial, contaminated in both directions
5	Device and 3-D Secure telemetry	Device fingerprint, IP, session data, authentication result	Real time	Exists only for card-not-present traffic; the gap is structural, not random
6	Network and consortium signals	Visa Advanced Authorization, Mastercard Decision Intelligence, compromised-account alerts	Real time to batch	Vendor black box the bank is accountable for but cannot inspect
7	Case management	Analyst dispositions, verification outcomes, rules fired	Continuous	Covers only already-flagged transactions, so today's rule bias becomes tomorrow's ground truth
8	Servicing logs	Travel notices, card locks, complaints	Real time to daily	Sparse and self-selecting, yet the only record of what a false decline costs

Note. MCC = merchant category code. PAN = primary account number. AVS = address verification service. TC40 and SAFE are the fraud reporting files issued by Visa and Mastercard.

Table 3***Preprocessing Steps and Their Failure Modes***

#	Step	What breaks if it is skipped
1	Validate schema, pair authorization to clearing, collapse reversals and terminal retries	Retries read as velocity attacks; reversals inflate spend history
2	Tokenize PAN (PCI Security Standards Council, 2024), link accounts across reissues, normalize merchant descriptors	A ten-year customer with a replaced card looks like a first-day risk
3	Store UTC, derive merchant-local time separately	Time-of-day features become noise for anyone who travels
4	Build the label from sources 4 and 7, set a maturity cutoff, mark declines as unlabeled	Declines are treated as legitimate, which is false
5	Flag missing AVS, CVV, 3DS and device data instead of imputing	Imputation destroys a real predictor, since the missingness describes the channel
6	Convert to base currency at transaction-date FX, log the amount, keep the raw amount	No way to compute fraud dollars prevented, a Section 1 success criterion
7	Build rolling velocity aggregates per card, merchant, device and IP; deviation from that card's own baseline; distance and time between consecutive transactions; encode MCC and merchant ID inside the training fold only	Encoding fitted on the full dataset leaks the label straight into a feature
8	Compute every aggregate as of the authorization timestamp, split out of time, weight classes on the training fold only, recalibrate afterward	The two classic failures: .99 offline and useless live, or a threshold set on probabilities that no longer mean anything

Note. FX = foreign exchange. Step 7 encoding refers to target or weight-of-evidence encoding of high-cardinality categories.

Steps 7 and 8 carry most of the work. *Customer spending behavior*, the input named in Section 1, is not a stored field. It is manufactured here, and it exists only if account linking in step 2 and the point-in-time logic in step 8 are correct.

The Label Problem

Source 4 is the weakest data in the system, and everything supervised depends on it. Fraud nobody notices stays labeled legitimate. A real purchase the cardholder later disputes gets labeled fraud. Both classes are contaminated,

in opposite directions.

The structural problem is worse. A declined transaction produces no purchase and no dispute, so the system never learns whether the decline was correct. Training data describes only the traffic the current model approved, and each retrain narrows that view (Lakkaraju et al., 2017). This lands on the Section 1 objective of avoiding unnecessary disruption: as built, the system cannot measure the harm it exists to prevent. False-positive rate is the one success criterion with no data behind it, and Section 6 carries it forward as a known limitation.

Table 4
Bias Entry Points and Safeguards

Entry point	What could happen	Safeguard
Merchant category and postal code	Strong predictors that also proxy for income and neighborhood, so disparities appear without any protected attribute in the model	Proxy-sensitivity testing; prefer less-disparate alternatives when the accuracy cost is small
Thin customer history	A new or recently reissued account has no personal baseline, so the model falls back on population priors	Offer a verification path before blocking; link accounts across reissues (step 2)
Location and device change	Travel or a new phone looks like fraud	Weigh other transaction details before treating a location or device change as a fraud signal
Card-present traffic	Device and 3DS signals are structurally missing, so some customers are always scored on thinner evidence	Track subgroup performance by channel; do not treat missingness as a risk score
Fraud label quality	Chargebacks can reflect billing disputes or processing errors rather than fraud	Confirm the investigation outcome before using a transaction as a fraud example
Decline feedback loop	A group over-declined today generates less counter-evidence tomorrow, so the disparity compounds at each retrain	Sample and review declined transactions; repeat the fairness audit after every retraining cycle
Analyst dispositions	Existing rule bias is laundered into the next training set as ground truth	Audit override patterns; record escalation reasons
Third-party scores	Network and vendor signals carry bias the issuer cannot inspect but is answerable for	Contractual transparency and vendor risk reassessment; monitor score distribution shifts
Aging training data	Older patterns miss both new customer behavior and new fraud methods	Retrain on a schedule; test whether older periods still contribute
Geographic concentration	The United States produced 26.31% of global card volume in 2024 but 41.87% of fraud losses (Nilson Report, 2026), so the model's baseline of normal is a U.S. baseline	Segment performance by region; do not port thresholds across markets

3. Model Selection and Validation Strategy

Model Families

Table 5
Candidate Model Families

Family	Role	Strengths	Limitations
Rules baseline	Benchmark and safety control	Transparent, operationally familiar	Cannot capture complex patterns
Logistic regression	Interpretable control	Simple, explainable, easy to calibrate	Misses nonlinear relationships
Random forest	Comparative benchmark	Robust to nonlinearity and interactions	Less efficient, often poorly calibrated
Gradient boosted trees (XGBoost, LightGBM)	Primary candidate	Strong on structured transaction data, fast at inference, handles imbalance with weighting	Overfits the minority class without careful tuning; needs calibration
Anomaly detection	Complementary signal	Catches previously unseen behavior	Anomalous is not the same as fraudulent
Sequence and neural models	Future option	Models complex behavioral sequences	Higher compute, harder to explain, heavier monitoring burden

Recommendation: gradient boosted trees as the primary model, with the rules engine retained as a benchmark and safety control, logistic regression as an interpretable reference, and anomaly detection as a complementary signal rather than a decision mechanism. Three criteria drive the choice, each tied to a Section 1 stakeholder priority. Inference is fast enough to score at checkout inside the latency budget, which serves merchants and cardholders. The precision/recall balance can be tuned to the operating point analysts and the bank actually need (Chen & Guestrin, 2016; Dal Pozzolo et al., 2018). And feature importance with SHAP values lets analysts and compliance see why a transaction was flagged (Lundberg & Lee, 2017), satisfying the explainability requirement.

Deep learning is deferred rather than rejected. It conflicts with the explainability and latency requirements today, and can be revisited if explanation tooling matures.

Validation Plan

Random k-fold cross-validation is not appropriate here. Random splitting lets future transactions and related customer behavior appear in training data used to evaluate earlier transactions, producing temporal, feature, and entity leakage (Kaufman et al., 2012). The plan instead is:

1. Sort all transactions chronologically.
2. Define a training window.
3. Insert a label-maturation buffer reflecting real chargeback and investigation timelines. A 30-to-90-day buffer is the starting proposal, to be confirmed against operational label data.
4. Define a validation window after the buffer.
5. Reserve a later, untouched test window.
6. Apply blocked walk-forward validation across several periods.

For example: fold 1 trains on months 1–6 and validates on month 8; fold 2 trains on months 1–7 and validates on month 9; and so on. Expanding windows retain all history; rolling windows keep only a recent period and adapt faster to concept drift (Gama et al., 2014). Resampling, class weighting, or synthetic minority generation applies to training data only. Validation and test sets keep the natural fraud prevalence, otherwise reported precision is meaningless.

Metrics. PR-AUC is the primary ranking metric, because it focuses on the minority class and the precision/recall trade-off (Saito & Rehmsmeier, 2015). Reported alongside it: precision, recall, false-positive rate, F-beta, fraud dollars prevented, expected net benefit combining fraud loss with review cost and customer impact, alert volume against analyst capacity, decision latency, calibration (Niculescu-Mizil & Caruana, 2005), and segment-level performance. ROC-AUC is a secondary diagnostic only; it can look strong while the model floods analysts with false alerts.

4. Ethical, Fairness, and Privacy Assessment

Fairness Target

Three fairness definitions were considered. Demographic parity, requiring similar approval rates across groups regardless of true risk, is a poor fit, since genuine fraud rates and transaction contexts do differ. Predictive parity, requiring similar actual-fraud rates among flagged transactions, supports analyst trust and is tracked as a secondary check. **Equalized odds is the primary target:** comparable false-positive rates on legitimate transactions and comparable recall on fraudulent ones across groups, with particular attention to false positives, which is where customer harm concentrates (Barocas et al., 2023; Verma & Rubin, 2018).

Unfairness enters through four channels already visible in Section 2: label bias from uneven historical investigation, proxy variables such as merchant category and geography, self-reinforcing feedback from differential scrutiny, and data sparsity in smaller subgroups.

Table 6
Subgroup Impact Analysis

Subgroup	Source of disparate treatment	Primary harm
Elderly customers	Less familiarity with app-based OTP or biometric verification; atypical transaction patterns	Failed step-up verification, declined legitimate purchases, reduced financial independence
Lower-income customers	Smaller transactions with greater variance resemble anomalies; fewer backup payment methods	A false decline disrupts essential purchases and creates financial stress
Recent immigrants	Cross-border transactions, remittances, unfamiliar merchants, thin credit history	Higher false-positive rates and more manual holds
Frequent international travelers	Transactions outside the normal geographic radius	Repeated flags damage trust and disrupt travel
Small and independent merchants	Merchant categories that statistically carry higher fraud rates	Lost sales and checkout friction, felt hardest by small businesses
Younger, thin-file customers	Short history gives less behavioral baseline	Higher false-positive and false-negative rates from data sparsity, not actual risk

Measurement. Compute subgroup confusion matrices covering false-positive rate, false-negative rate, and time to resolution for held transactions. The proposed investigation trigger is that no subgroup's false-positive rate exceed 1.5 times the population rate. As an illustration: if the population rate is 2% and recent-immigrant customers experience 5%, the 2.5× disparity is above threshold, and the correct response is root-cause analysis, such as testing whether recurring remittance patterns are being misread, rather than reflexively deleting geography as a feature. Audits repeat after every retraining cycle, since feedback loops widen disparities over time (Chouldechova, 2017).

Mitigation. Build subgroup-specific baselines for legitimate-but-atypical patterns such as remittances and travel. Widen the middle tier so borderline transactions route to step-up verification instead of an immediate decline, with an accessible alternative such as phone verification. Apply proxy-aware feature governance. Provide a fast appeal path and feed validated appeal outcomes back into training. Maintain a subgroup fairness dashboard with defined escalation, and commission an independent audit at least annually.

Privacy and Governance

Transaction records reveal where, when, and how often a person buys, which makes over-collection the central privacy risk alongside breach exposure (National Institute of Standards and Technology [NIST], 2020). Controls: data minimization and purpose limitation, tokenization or pseudonymization of identifiers in development datasets, role-based access under least privilege, encryption in transit and at rest, documented retention and deletion schedules, and an explicit prohibition on secondary use. Protected attributes are excluded from production decisioning and retained separately, under restricted access, solely for fairness testing. Institutions covered by the FTC Safeguards Rule must also ensure their service providers protect customer information (Federal Trade Commission, n.d.), which extends these obligations to payment networks, cloud platforms, and fraud vendors.

Governance follows model risk management practice (Board of Governors of the Federal Reserve System & Office of the Comptroller of the Currency, 2011; NIST, 2023). The institution names who develops, validates, approves, deploys, and monitors the model. Threshold changes follow documented review rather than informal adjustment. Maintained artifacts: a model card describing intended use and limitations, a feature register with data lineage, validation results and approval evidence, threshold and override documentation, and audit logs sufficient to reconstruct which model version and rules evaluated any given transaction. Analysts receive understandable alert reasons without exposing model logic that fraudsters could exploit.

5. Monitoring, Drift Detection, and Post-Deployment Plan

Figure 1 (the Fraud Sentinel dashboard mock-up, submitted as the group's visual artifact) shows the deployed system. The top strip traces the real-time path a transaction follows: data capture, feature lookup, fraud scoring against a sub-200ms target, a routed decision, and outcome logging. Below it, each headline metric is tied to the stakeholder who depends on it, which prevents the team from optimizing one group's outcome at another's expense: precision for analysts, recall for fraud loss, false-positive rate for customers, latency for merchants.

Three tracks form the workflow.

Drift. Population Stability Index monitoring on transaction amount, merchant mix, geography, device, and channel distribution. A PSI above 0.25 automatically pauses retraining and opens a review ticket (Lin, 2017; Yurdakul, 2018).

Error. False positives and false negatives, tracked in aggregate and by subgroup, with fairness deviations routed to compliance. Because labels mature slowly, monitoring runs in two stages: immediate operational signals such as alert volume, authentication success, and complaint rate, then confirmed fraud outcomes once the maturation window closes.

Human oversight. A defined escalation chain from automated flag to analyst triage to ML developer root-cause review to compliance sign-off, with customer-service feedback closing the loop. No model change deploys without human review.

Routing is risk-based. Low risk approves automatically. Medium risk requests additional authentication. High risk holds or declines and routes to an analyst, who can contact the customer and override the model. A \$2,000 purchase abroad that trips location, amount, and pattern checks is exactly the case that belongs with a human rather than an automatic decline.

Table 7
Escalation Triggers and Responses

Trigger	Response
Material decline in PR-AUC or recall	Investigate cause before acting; check data quality, drift, and system changes
Significant increase in false declines	Adjust thresholds; widen step-up verification tier
PSI above 0.25 on a monitored feature	Pause retraining, open review ticket
Alert volume exceeds analyst capacity	Raise review threshold temporarily; rebalance queues
Material subgroup disparity	Root-cause analysis, fairness re-audit, feature review
Feature outage or data-quality failure	Fall back to rules engine; remediate feature
Evidence of a new fraud pattern	Targeted retrain, or temporary rule to cover the gap

Retraining is never automatic on a performance drop. The sequence is investigate cause, collect recent data, retrain, validate on a future period, review for fairness and privacy, obtain approval, deploy, and resume monitoring. A champion-challenger arrangement runs the new model in shadow mode before it takes over production decisions, and rollback capability is retained.

6. System Redesign Proposal

The redesign treats fraud detection as an operational decision system rather than a classification algorithm. The target-state architecture has eight layers:

1. **Data ingestion** — transaction, customer, merchant, device, fraud-outcome, and operational data with documented ownership and lineage.
2. **Data quality and privacy** — validation, access control, retention and privacy enforcement.
3. **Point-in-time feature layer** — behavioral and historical features built only from information available at the moment of the transaction, served by one code path shared between training and inference.
4. **Modeling** — rules benchmark, interpretable baseline, gradient boosted trees as primary, optional anomaly signal.
5. **Risk scoring** — a calibrated score with model version, timestamp, and explanation attached.
6. **Decision** — business-approved thresholds for approve, authenticate, review, or decline.
7. **Human review** — uncertain and high-impact cases routed to analysts, with outcomes captured for future evaluation.
8. **Monitoring and governance** — drift, performance, fairness, privacy, workload, latency, and availability.

Implementation proceeds in six phases: baseline and audit of current rules and costs; leakage-controlled model development using walk-forward validation; threshold and workflow design driven by financial cost, analyst capacity, and customer impact; governance testing across fairness, privacy, explainability, and security; shadow deployment

alongside the existing system without changing customer outcomes; and controlled production release with rollback retained and scheduled post-deployment review.

Approval requires reliable future-period performance, acceptable customer and analyst impact, working human oversight, and no material unresolved leakage, privacy, or fairness risk.

One limitation carries forward unresolved. The system cannot observe outcomes for transactions it declined, so false-positive rate, the metric closest to the Section 1 objective, has no ground truth behind it. This is documented as a known constraint and managed through subgroup monitoring, the appeals path, and analyst review, rather than presented as something model choice can fix. A partial mitigation worth piloting is releasing a small randomized sample of borderline declines to observe their true outcomes, accepting a measured fraud cost in exchange for the counterfactual data the system otherwise never gets.

The insight we would keep. The hardest problems in a deployed ML system are not in the model. They are in the label, the split, and the feedback loop. A fraud model that reports .99 offline and fails in production usually failed at step 7 or 8 of the preprocessing table, not in the choice of algorithm.

Group Reflection

Dividing the work by lifecycle stage surfaced something none of us saw individually: the sections were not independent. Section 1 named false-positive rate as a success criterion, and Section 2 found that the data to compute it does not exist, because declined transactions never produce an outcome. Section 3's validation design only works because Section 2 defined a label-maturation buffer, and Section 4's fairness audit depends on the same declined-transaction blind spot that Section 2 flagged. Each dependency turned up when one member's finding contradicted an assumption in someone else's section, which is the part of the process we would not have reached working alone.

The main disagreement was how much to promise. Early drafts described the redesigned system as solving fairness and monitoring problems. We settled on documenting the declined-transaction gap as an open limitation instead, because claiming a fix we could not measure would have repeated the exact failure the assignment asks us to look for. If we ran this again, we would begin with the label and the evaluation plan rather than the model, since almost every downstream constraint traced back to those two decisions.

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